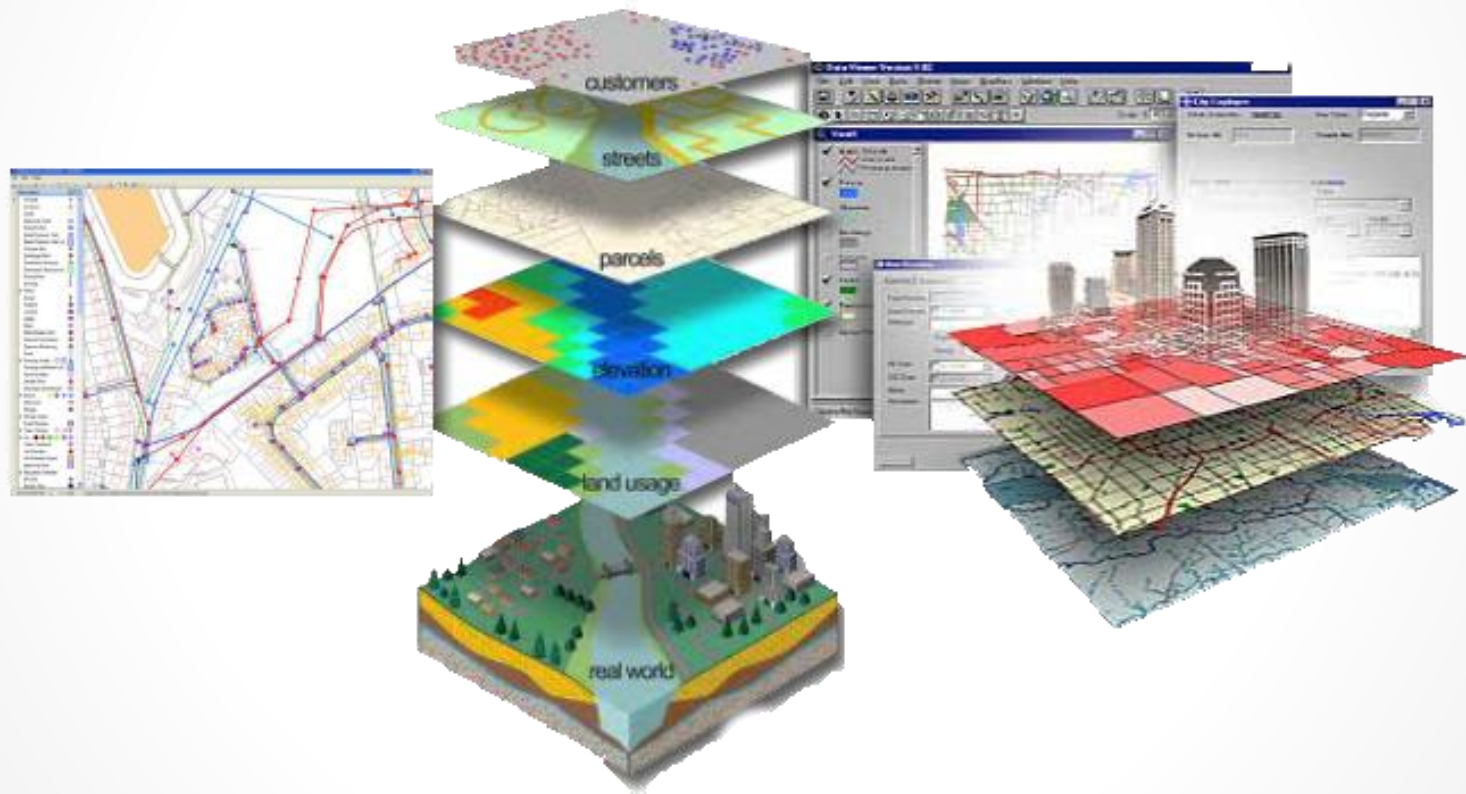


Geographic Information System

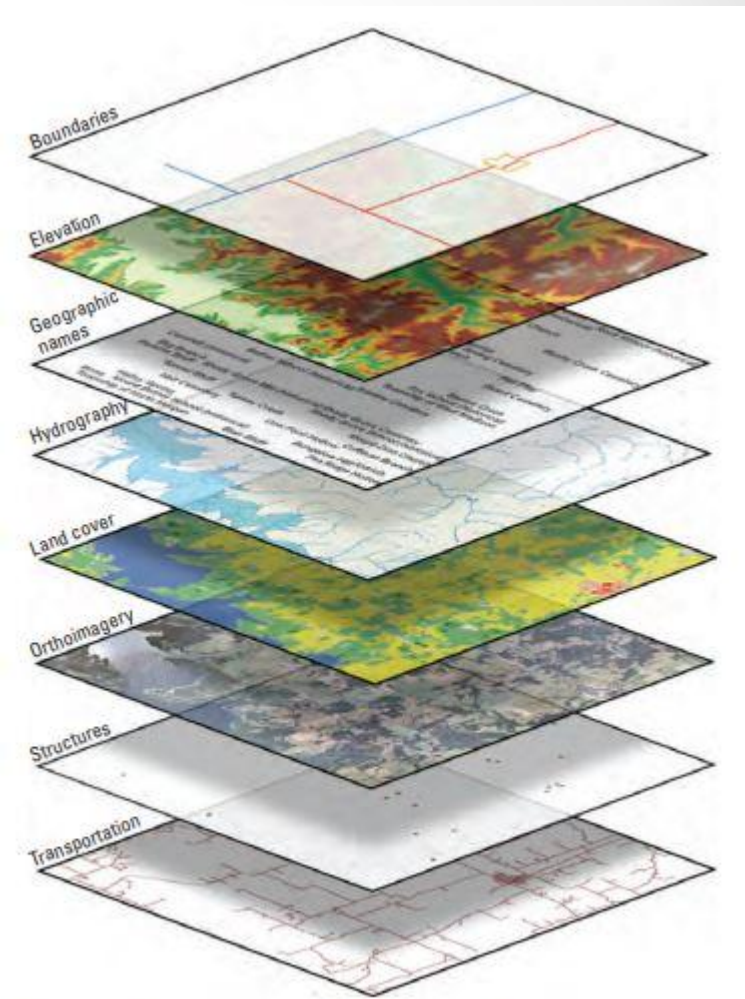
MND 409



MODULE 1

What is a geographic information system (GIS)?

A Geographic Information System (GIS) is a computer system that analyzes and displays geographically referenced information. It uses data that is attached to a unique location.



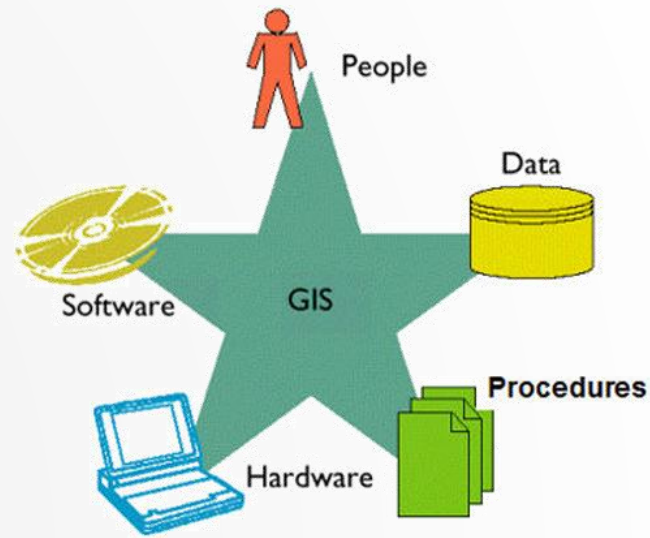
Eight base layers of *The National Map*.

Geographic Information System

- A Geographical Information System is a tool, which can accept large volumes of spatial data derived from a variety of sources, retrieve, manipulate, analyse and display them according to user-defined specifications
- A type of Decision Support System

Components of a GIS:

- Hardware
- Software
- Spatial Data
- Procedures
- People

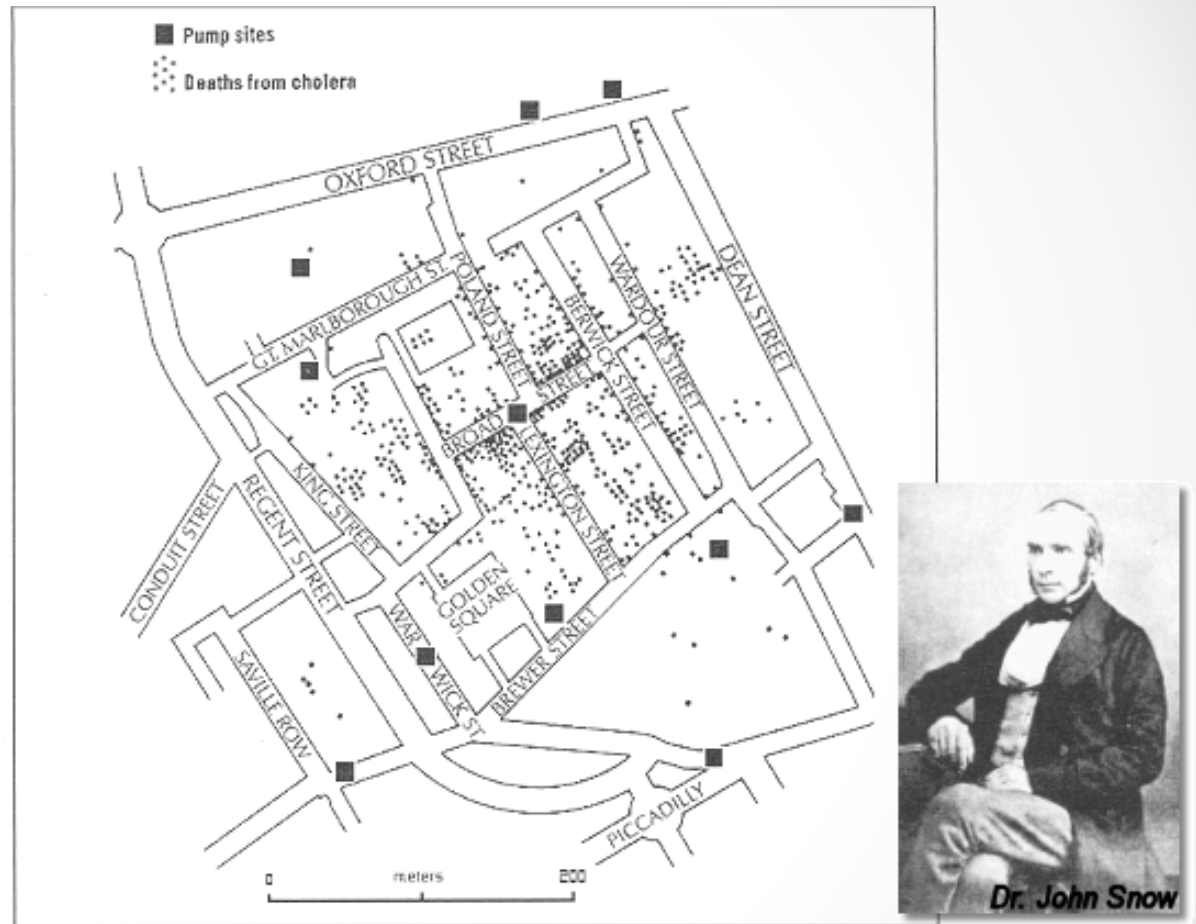


History of GIS

- Basic, unscaled maps have been used to depict small regions/area from very ancient times
- Slowly, art of *cartography** evolved.... People started using maps for marking administrative boundaries and topography
- *Thematic Maps** were generated to portray spatial information of surface features like soils, land-use, geology, census, rainfall, natural resources distribution, vegetation,
- Much more geographic information is conveyed when multiple layers of data are super-imposing over a base map
Eg: Dr. John Snow mapped cholera deaths and pump-locations over the map of London to find the cause of the disease outbreak

Year	Milestone	Description
	The first maps	The roots of GIS go back hundreds, even thousands of years in the fields of cartography and mapping. Early maps are used for exploration, strategy, and planning.
1854	Early spatial analysis	In London, physician John Snow maps Cholera cases to trace the origin back to one water source.
1960	Geographic computer science emerges	Throughout the 1960s, computers advance significantly in technology, speed, and design, with IBM leading the way. Early concepts of quantitative and computational geography begin to develop.
1963	The first GIS	Geographer Roger Tomlinson begins a national land use management program for the Canadian government, inventorying natural resources. Tomlinson first coins the term geographic information system (GIS) and becomes known as the <i>"father of GIS"</i> .

History of GIS



(http://healthcybermap.org/HGeo/pg1_1.htm)

Year	Milestone	Description
1965	Harvard Lab explores spatial analysis	Architect Howard Fisher establishes the Harvard Laboratory for Computer Graphics and Spatial Analysis, where some of the first GIS software, such as ODYSSEY, is invented and computer mapping applications explored. GIS pioneers such as Jack Dangermond, Carl Steinitz, Scott Morehouse, Allan Schmidt, and Allen Bernholtz participate.
1969	Esri is founded	With inspiration from the Harvard Lab and <i>Design with Nature</i> , Jack and Laura Dangermond form Environmental Systems Research Institute (E.S.R.I.), now known as Esri. They begin project work, helping land use planners make better decisions with an emphasis on protecting the environment.
1972	First Landsat satellite	Inspired by photographs taken from space during NASA's Gemini IV mission, the US government launches the first of many Landsat satellites for Earth observation. The program provides current satellite imagery of the whole world, tied to geographic points. This becomes a major data input for GIS and begins the era of remote sensing—changing how we see the Earth.

Year	Milestone	Description
1978	First GPS satellite	The US launches its first GPS satellite, Navstar I, and achieves full global coverage with GPS in 1994. Other nations develop similar Global Navigation Satellite Systems (GNSS). With global coverage, virtually any object can report its position, and the amount of spatial data begins to grow exponentially.
1982	GRASS develops	US Army Corps of Engineers begins development of GRASS (Geographic Resources Analysis Support System). This open-source GIS software broadens access to GIS and is still available today.
1986	In-car navigation	Etak develops the first in-car navigation concepts. Mazda's Eunos Cosmo is the first car with an automatic navigation system. Later, real-time GIS capabilities will enable the possibility of self-driving cars from companies such as Tesla.

Year	Milestone	Description
1991	GIS goes mainstream	GIS experiences a turning point, becoming crucial to many workflows. Publications like Forbes, Fortune, and Business Week start writing about GIS.
1999 (18 November)	First GIS day	Jack Dangermond, Dr. Roger Tomlinson, and students celebrate the first GIS Day at Murch Elementary School in Washington, DC. Dangermond credits Ralph Nader as the person who inspired GIS Day, now celebrated worldwide on the third Wednesday in November during Geography Awareness Week.
2004	National Geospatial Agency (NGA)	In the United States, the National Imagery and Mapping Agency (NIMA) is renamed to NGA to emphasize the growing importance of geospatial intelligence, reflecting a broader convergence of imagery, mapping, and other geospatial information in GIS.

Year	Milestone	Description
2005	Google Maps	With the advent of mapping apps like Google Maps and Google Earth, everyone can now interact with and benefit from GIS technology and it begins to become embedded in our everyday lives.
2006 - 2018	Cloud computing, First iPhone, Digital data libraries, ArcGIS online, AI & GIS	Jack Dangermond, Dr. Roger Tomlinson, and students celebrate the first GIS Day at Murch Elementary School in Washington, DC. Dangermond credits Ralph Nader as the person who inspired GIS Day, now celebrated worldwide on the third Wednesday in November during Geography Awareness Week.
2020	GIS supports COVID-19 response	Johns Hopkins University creates an ArcGIS-powered COVID-19 tracking dashboard that receives over a trillion views. The online dashboard becomes the go-to resource for monitoring the global health crisis. It inspires thousands of similar dashboards and helps people understand the pandemic, bringing GIS to the forefront of public knowledge.

History of GIS

- Before GIS, all these maps were recorded on paper or film media for their use and storage.

Limitations of 'Paper' based maps:

- Frequent updating is difficult
 - Analysis especially while integrating multiple data-sources
 - Changes/Updation in one map is not reflected in all other copies of the same map
 - Paper media is subjected to 'wear and tear' damages
- Slowly people realised these limitations of 'film or paper' based maps and a strong need of GIS was felt.

History of GIS

Computer based GIS started in 1960s.

Stages of Evolution:

Pioneering phase 1960 – 1973

- Individual personalities played the role in determining what could be achieved

Second phase 1973 – 1980

- Involvement by national agencies in various parts of the world to put GIS into practice (eg. GSI in India)

Third phase 1980 – 1990

- Commercial dominance

Current Phase

- User dominance bringing standardization, access across network, interoperability, OGIS formation etc

GIS in India

- Adoption of GIS in India started in late 1980's with Natural Resources Data Management System (NRDMS), National Natural Resources Management System (NNRMS), and National Natural Resources Management System (NNRMS).
- Urban Mapping Scheme (UMS) which involved base map preparation using aerial photographs for 53 towns as pilot scheme, which was continued under National Urban Information System (NUIS) scheme in tenth five-year plan.
- 1990s- Indian Defence, India Census, Integrated Mission for Sustainable Development, National (Natural) Resources Management System (NRIS), adopted GIS during this period. In the private sector CESC and Reliance Infocomm, commenced deployment of electric GIS and Telecom GIS respectively. Satellite imageries were used extensively for generating telecom propagation models for planning by the private telecom industry.

GIS in India

- 2000s- Use of GIS became prominent in governance. Many government schemes were formulated with GIS as integral component of the implementation plan. These include Restructured Accelerated Power Development and Reforms Programme (R-APDRP), The Digital India Land Record Modernization Programme (DILRMP), previously known as the National Land Record Modernization Programme (NLRMP), The Jawaharlal Nehru National Urban Renewal Mission (JNNURM), Integrated Watershed Management Program (IWMP), and many others.
- This was the decade which also saw massive geospatial infrastructure deployment in the country. These include National Forest Information System (NFIS), BharatMaps- Multi-Layer Geographical Information System (GIS) for Planning - NICMAPS, National Urban Information System (NUIS), National Spatial Data Infrastructure (NSDI), and other state geospatial infrastructures.

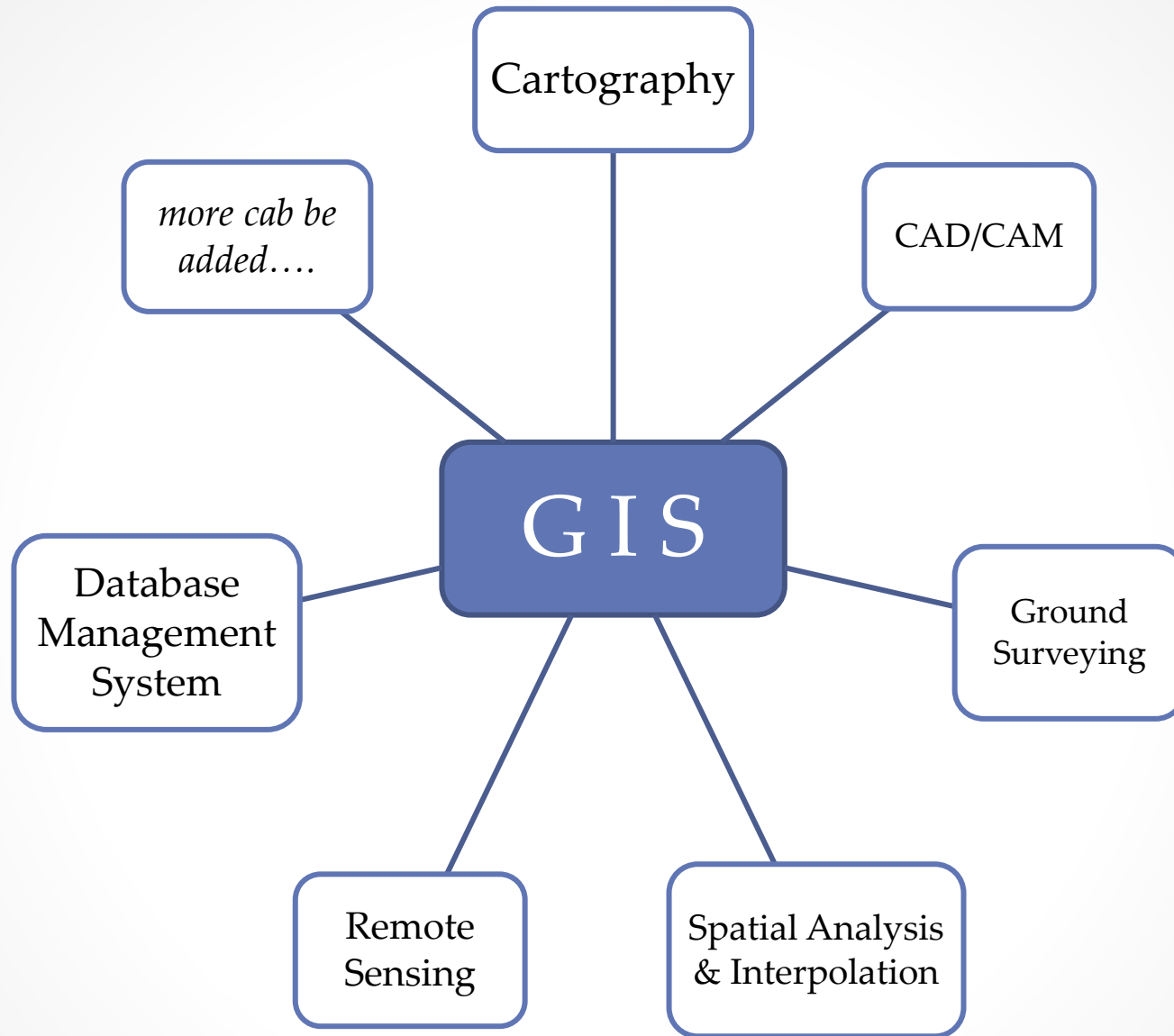
GIS in India

- 2010s- Decade of mass adoption with use of GIS spreading across the economic activity in the country. There was a surge of GIS deployment in the country for addressing various challenges covering infrastructure, water resource management, rural development, urban development, public safety, healthcare, agriculture, environment and natural resource management and disaster management. This decade also saw strengthening of geospatial infrastructure in terms of Continuously Operating Reference Station (CORS) networks being deployed in the country along with further expansion of GIS based decision support systems and state spatial data infrastructures.
- 2022- India's National Geospatial Policy on December 28, The **National Geospatial Policy, 2022**, notified by the Government of India on **December 28, 2022**, is a transformative policy aimed at positioning India as a global leader in the **geospatial sector**. With a long-term vision extending to **2035**, the policy seeks to **liberalize** and **democratize access** to geospatial data, fostering innovation and enabling its widespread use across governance, businesses, and academia.

spatial ('speɪʃəl)

adj

1. of or relating to space
2. existing or happening in space



• GIS borrows features from separate but closely related fields •

GIS Applications

- Resources Management
- Land-use Planning
- Agriculture
- Forestry
- Water Resources Management
- Rural/Urban Planning
- Environmental Management
- Risk Management
- Business /Marketing
- Real Estate
- Facility Mapping
- Transportations Management
- Telecommunication
- Mining
- Government Agencies
- Defence
- Emergency Operations
- Crime Management
- Epidemiology
- Archaeology
-
-
-

Data Source in GIS

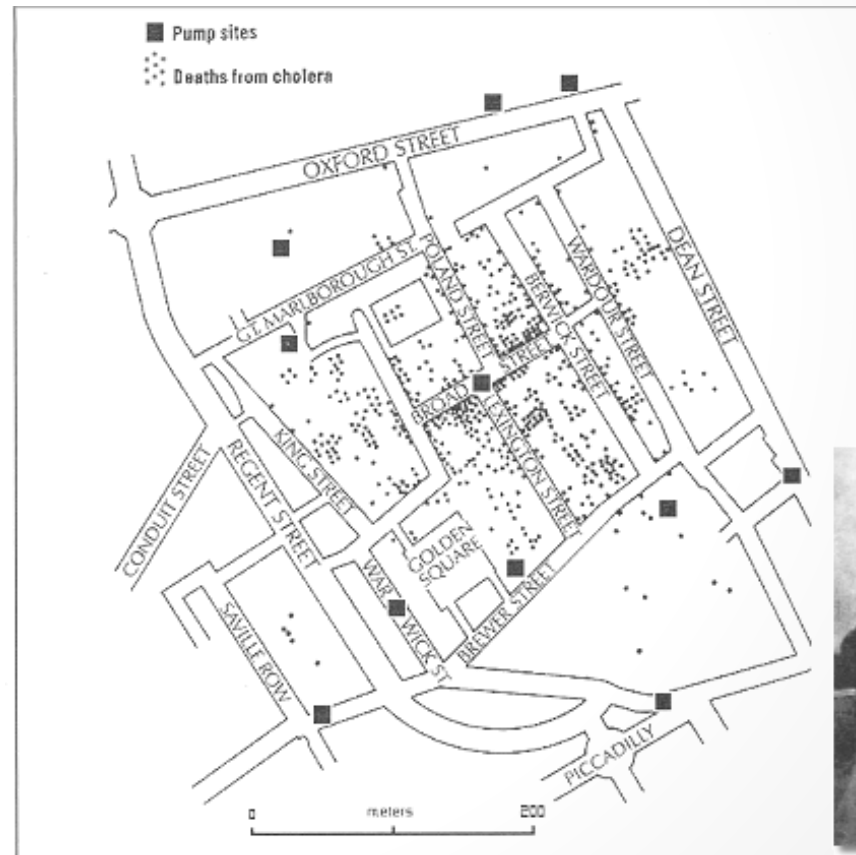
- Maps: Cadastral Maps, Topographical Maps and Thematic Maps
- Aerial Photographs & Satellite Imageries
- Ground Surveys
- GPS
- Government Agencies (like Geological Survey of India)
-
-
-

Maps

A map is a two dimensional small scaled representation of earth surface (or it's part) as seen from above

- It shows the *spatial relationship* between a specific feature or phenomena
- Map is the assemblage of four categories of information:

Point, Line, Area & Label

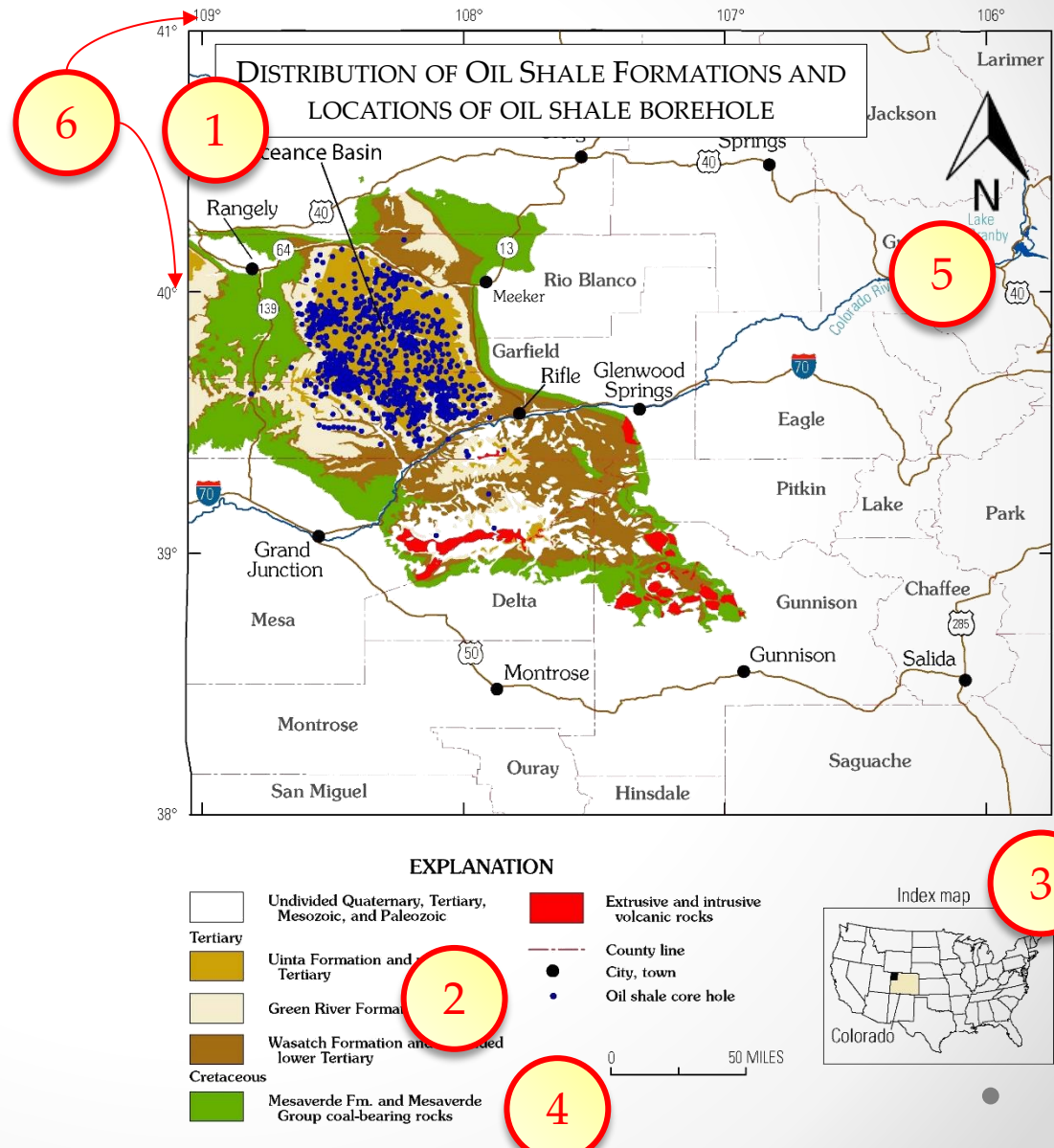


Dr. John Snow

(http://healthcybermap.org/HGeo/pg1_1.htm)

Basic elements of a Map

1. Title
2. Legend
3. Index
4. Scale
5. Direction
6. Grid System



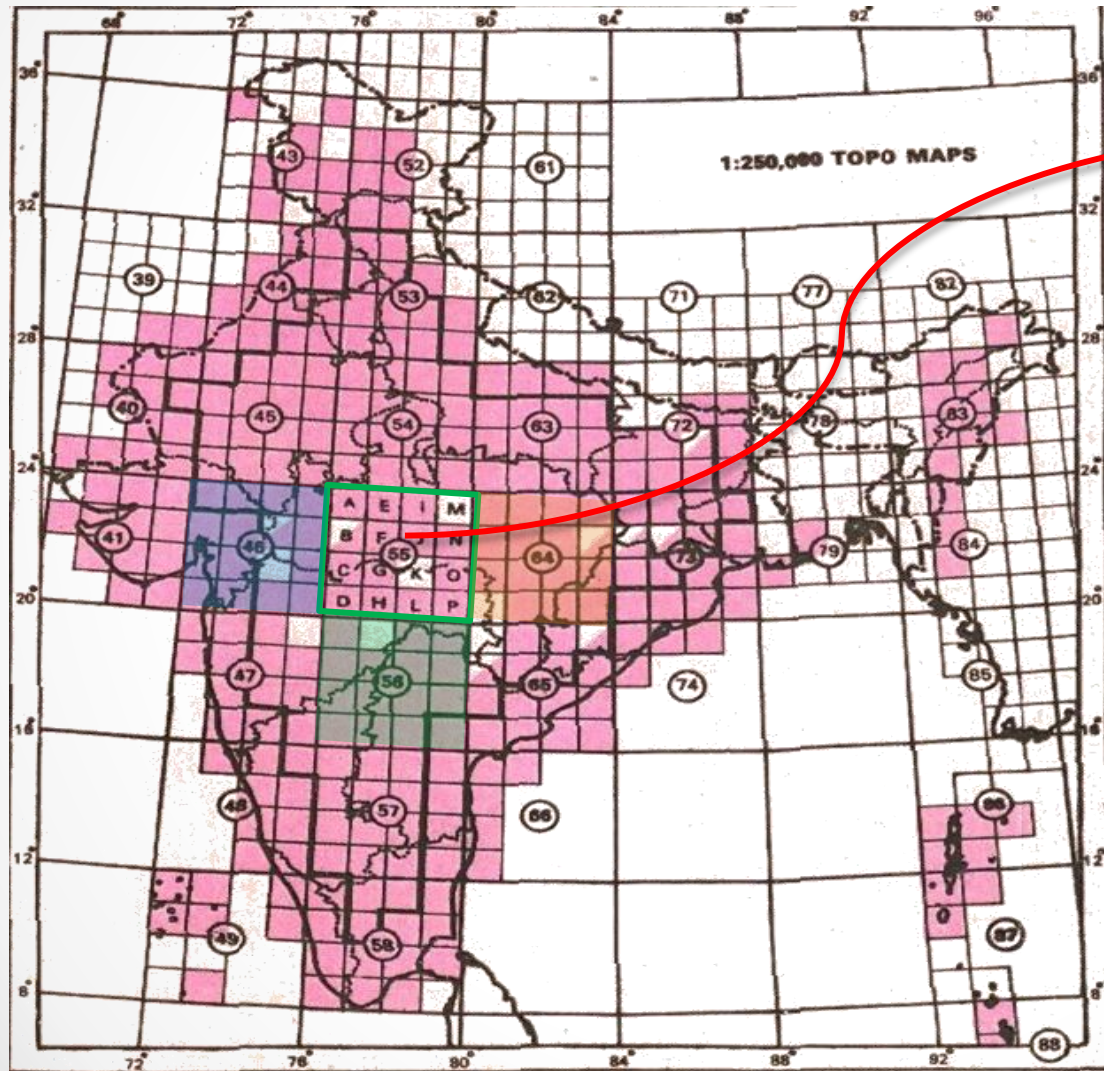
Map Scale

Scale is written as a fraction called
Representative Fraction

$$R.F. = \frac{\text{Distance on Map}}{\text{Distance on Ground}}$$

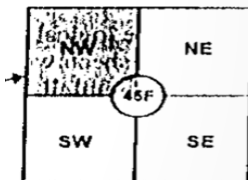


Topographical Maps Of India



45 F : degree sheets 1° x 1°
1: 1,250,000

A	E	I	M
B	F	J	N
C	G	K	O
D	H	L	P



45 F/6 : Inch map (15' x 15')
1: 50,000

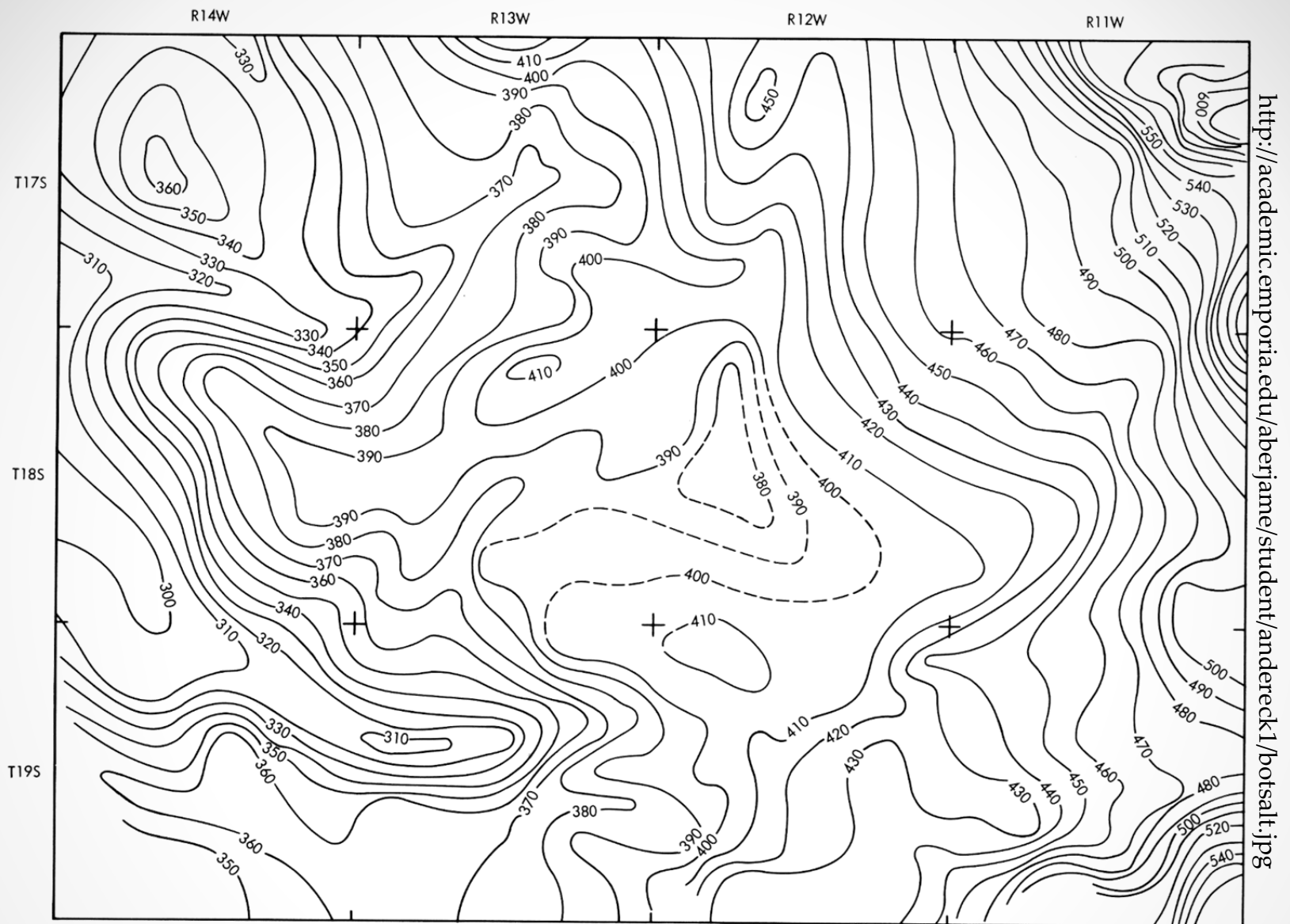
1	5	9	13
2	6	10	14
3	7	11	15
4	8	12	16

Thematic Maps

These Maps represent the spatial distribution of a particular phenomena (theme) of our interest

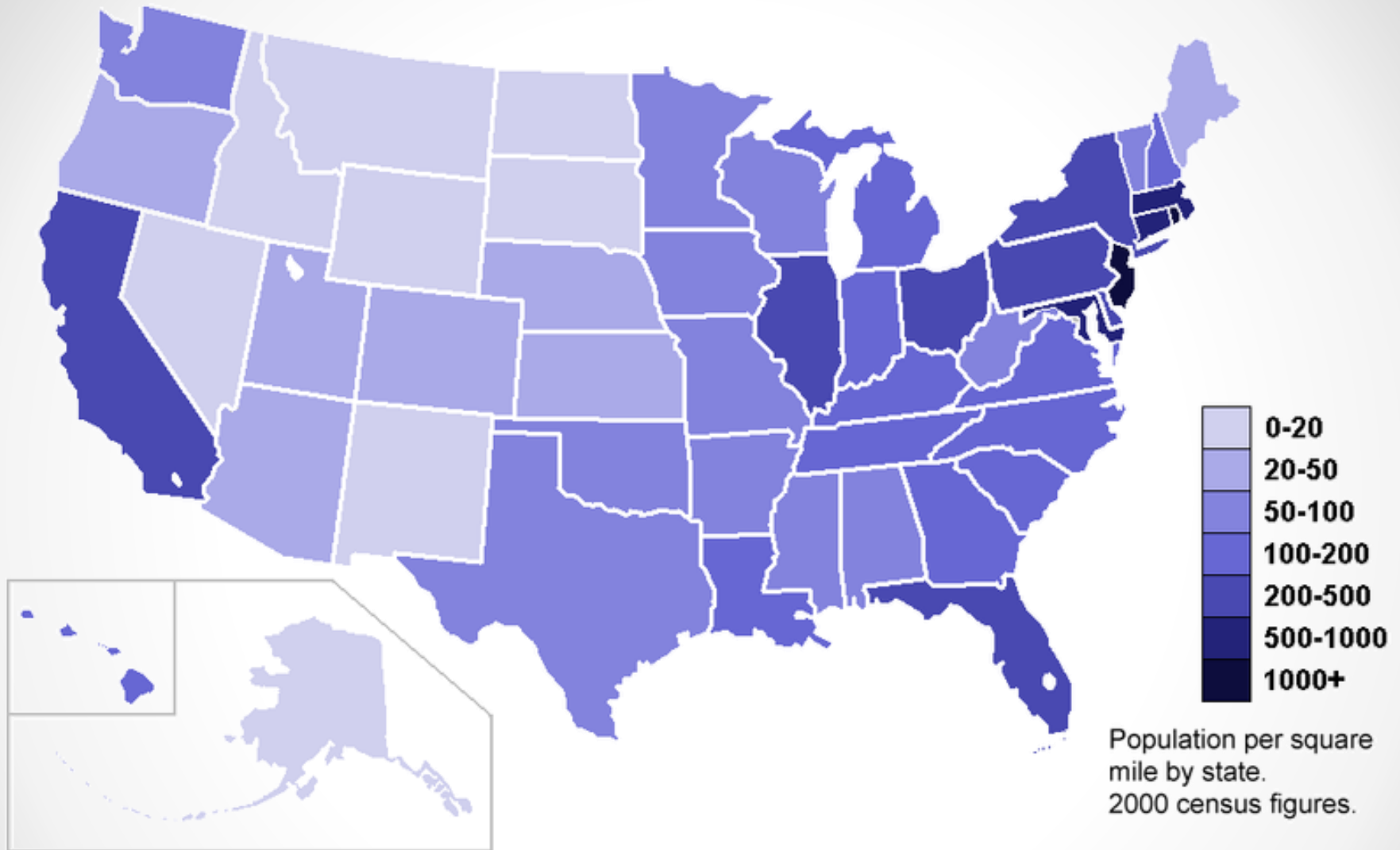
Mapping methods used for creating a thematic map:

1. Isopleth
2. Choropleth
3. Other Graphical Representations



Isopleth has the iso-line connecting points of equal magnitude

Eg: Contour Map, isotherm, isobar....



Choropleth has tonal shading/ colour variations to represent statistical data aggregate over predefined region